



Unit III – Random Processes

Basics · Classification · Stationarity · Autocorrelation

Engineering Mathematics V – Probability & Random Processes

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Lecture 25 – Random Process Basics

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1 What is a random process?

Random Process (RP)

A **random process** is a *collection of random variables* (or a *collection of random signals*).

- A **random variable** involves **only** the sample space S : $X = X(s)$, $s \in S$.
- A **random process** involves **both** the sample space S and a parameter (time) set T : $X = X(t, s)$, $t \in T$, $s \in S$.

Usually the argument s is suppressed and we simply write $\{X(t)\}$.

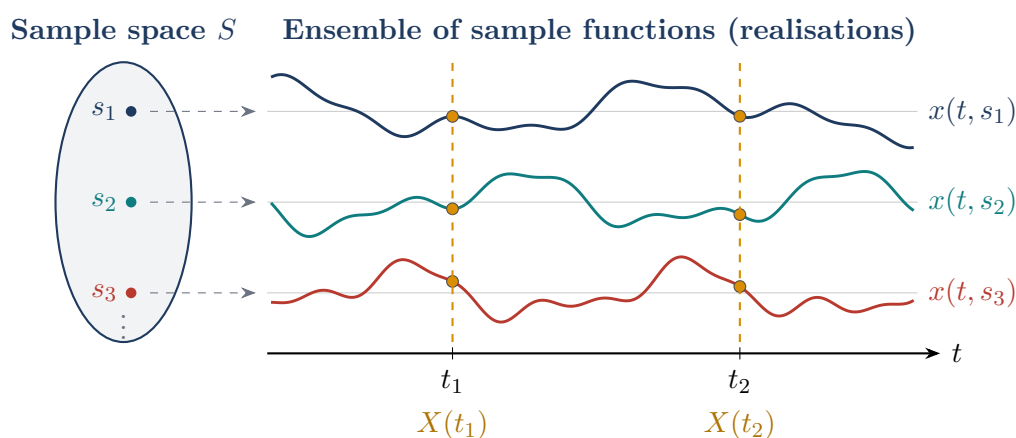
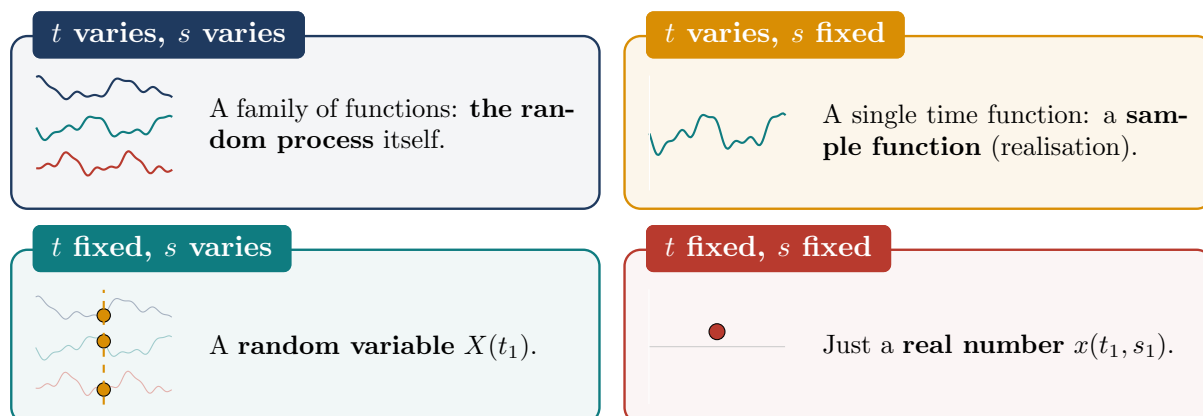


Figure 1: A random process as a collection of signals. Fix t and look across the ensemble: each vertical slice is a random variable.

1.1 Four ways to read $X(t, s)$



2 Two motivating examples

2.1 Example 1: arrivals at a railway junction

Example 1

$X(t)$ = number of people arriving at a railway junction in the time interval $(0, t)$ hours.

- $t = 2$: $X(2)$ is one random variable,
- $t = 3.5$: $X(3.5)$ is another random variable,
- $t = 10$: $X(10)$ is yet another random variable.

This number of arrivals may be represented through $(1 + at)^n$; e.g.

$$P\{X(t) = n\} = \frac{(at)^{n-1}}{(1 + at)^{n+1}}, \quad n = 1, 2, 3, \dots$$

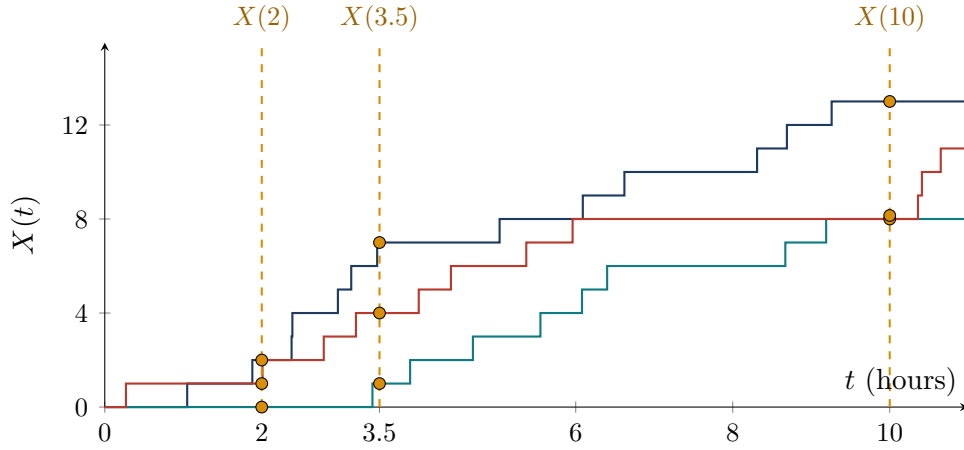


Figure 2: Three sample functions of the arrival process. At each fixed time instant the ensemble of values forms a random variable.

2.2 Example 2: signals of a radio station

Example 2

Signals of a radio station received by different receivers, lying at different distances, may be represented as a random process

$$X(t) = A \cos(\omega t + \phi).$$

Here the amplitude A and the phase ϕ vary with the distance of the receiver from the radio station; hence A and ϕ are random variables (similarly t may also be treated as a random variable).

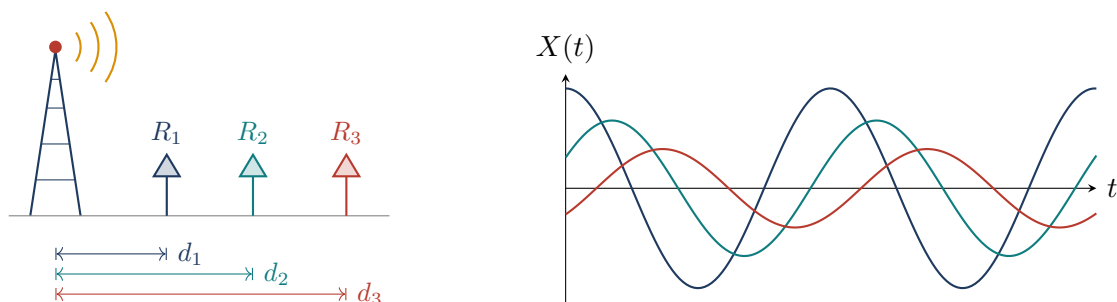


Figure 3: Receivers R_1, R_2, R_3 at distances $d_1 < d_2 < d_3$ (left) and the signals they receive (right): a farther receiver gives a different amplitude A and phase ϕ .

3 Classification of random processes

Based on the nature of the parameter set T and the sample (state) space S , we have the following classification of random processes.

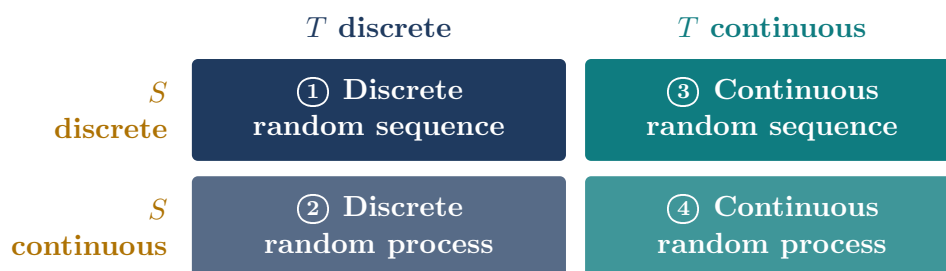


Figure 4: The four types of random processes.

Memory aid

T decides **discrete** / **continuous**; S decides **sequence** / **process**.

3.1 Type 1: both T and S discrete

Discrete random sequence

If both T and S are discrete, the RP $X(t, s)$ is called a **discrete random sequence**.

Example

Number of child births in the time interval of every one hour, in a metropolitan city.

$$S = \{0, 1, 2, 3, 4, \dots\}, \quad T = \{1, 2, 3, \dots, 24\}.$$

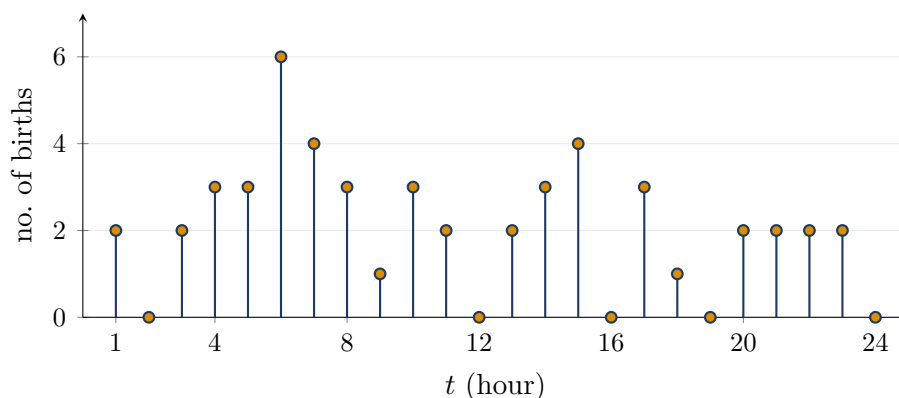


Figure 5: Type 1: isolated time points, integer values.

3.2 Type 2: T discrete and S continuous

Discrete random process

If T is discrete and S is continuous, the RP $X(t, s)$ is called a **discrete random process**.

Example

The temperature recorded at a place in every 4 hours of duration.

$$S = \{15.8^\circ\text{C} - 40.6^\circ\text{C}\}, \quad T = \{4, 8, 12, 16, 20, 24\}.$$

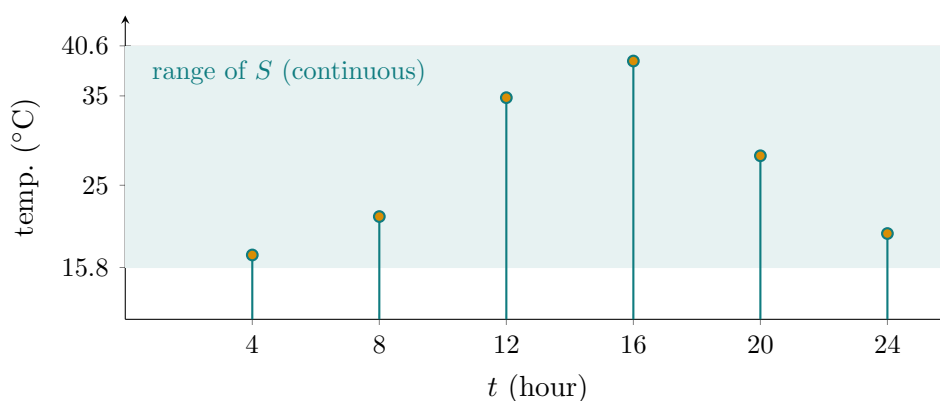


Figure 6: Type 2: isolated time points, any real value in a range.

3.3 Type 3: T continuous and S discrete

Continuous random sequence

If T is continuous and S is discrete, the RP $X(t, s)$ is called a **continuous random sequence**.

Example

Number of customers arriving at a counter in the time interval of $(0, t)$ minutes; $S = \{0, 1, 2, \dots\}$, $T = [0, \infty)$.

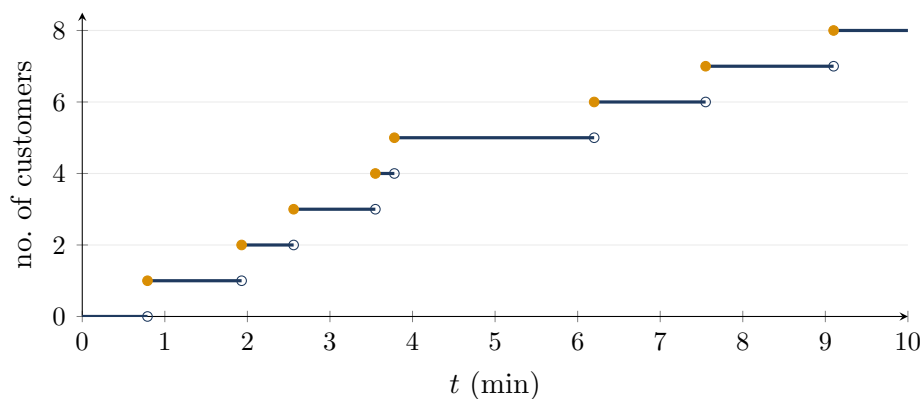


Figure 7: Type 3: defined for every t , jumps by integer steps.

3.4 Type 4: both T and S continuous

Continuous random process

If both T and S are continuous, the RP $X(t, s)$ is called a **continuous random process**.

Example

Maximum temperature recorded at a place in the time interval of $(0, t)$ hours; S is an interval of real numbers and $T = [0, \infty)$.

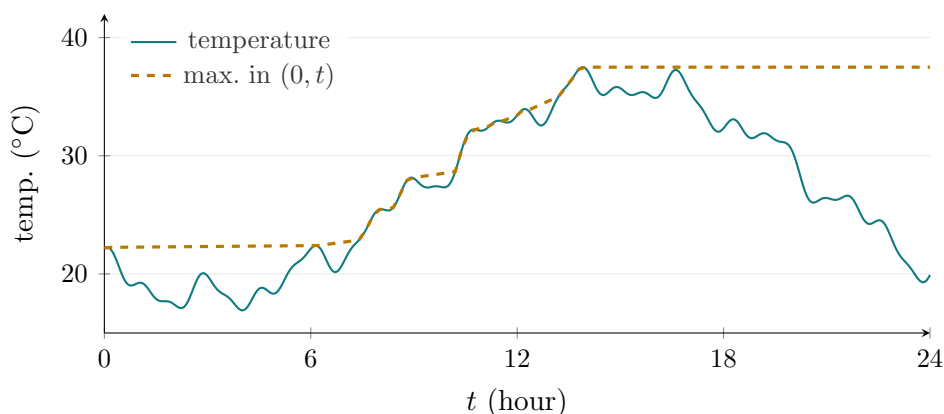


Figure 8: Type 4: defined for every t and taking any real value; the dashed curve is the maximum temperature recorded in $(0, t)$.

4 Stationary random processes

Stationary RP

A random process $X(t, s)$ whose distributions (values) do not vary with respect to the time parameter t is called a **stationary** random process.

Strictly stationary RP

A random process $\{X(t)\}$ is said to be **strictly stationary** if it does not vary with respect to the time t .

How do we check?

Compute the **statistical averages** of $X(t)$ – such as the **mean** and the **variance** – and see whether they depend on t .

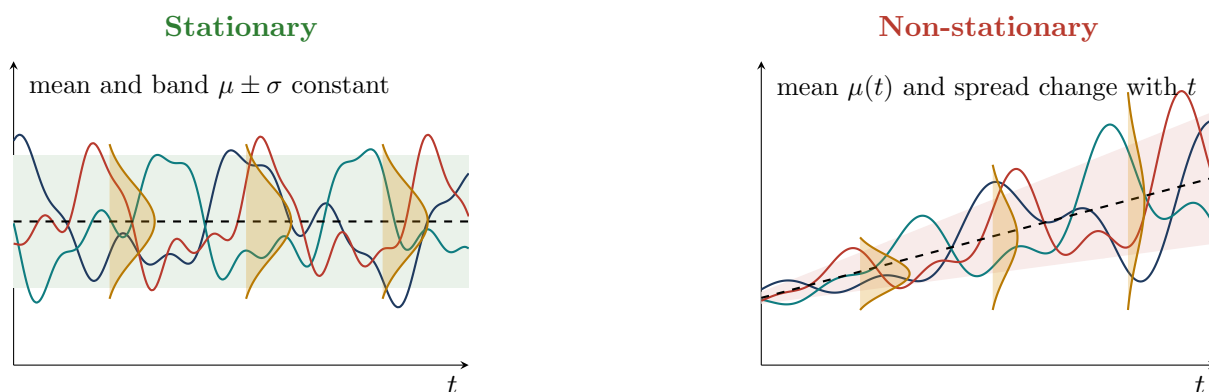


Figure 9: Amber curves: the distribution of $X(t)$ at three time instants – identical (left), shifting and spreading (right).

4.1 Strict-sense stationary (SSS) process

SSS / strongly stationary process

If both the mean and the variance values of a random process $\{X(t)\}$ are constant, then it is called a **strict-sense stationary** (SSS) or **strongly stationary** process. That is, $\{X(t)\}$ is an SSS process if

- (i) mean of $X(t) = \mathbb{E}[X(t)] = \text{a constant}$,
- (ii) $\text{Var}[X(t)] = \text{a constant}$,

i.e. neither is a function of time t .

Remark (formal view)

Formally, strict-sense stationarity means that *every* joint distribution of $X(t_1), X(t_2), \dots, X(t_n)$ is unchanged when all the time instants are shifted by the

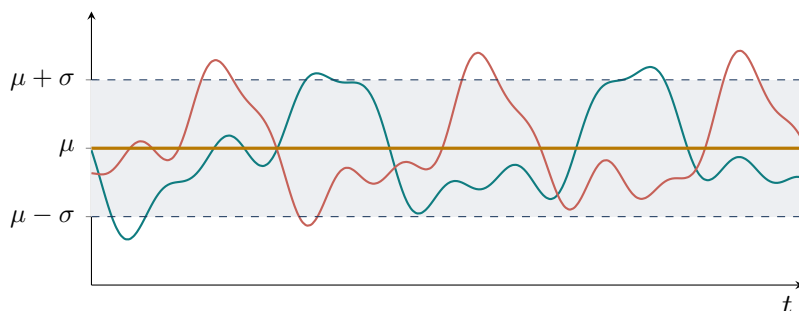


Figure 10: For an SSS process the mean line and the $\pm\sigma$ band are flat.

same amount c :

$$F(x_1, \dots, x_n; t_1 + c, \dots, t_n + c) = F(x_1, \dots, x_n; t_1, \dots, t_n).$$

Constant mean and constant variance are the working checks used while solving problems.

5 Correlation and the autocorrelation function

Correlation between two random variables

If X and Y are two random variables, then the **correlation** between X and Y is given by $\mathbb{E}(XY)$.

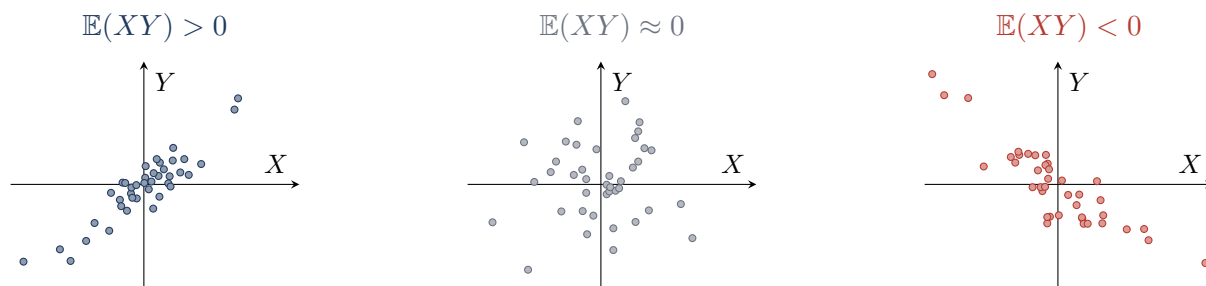


Figure 11: For zero-mean X and Y , $\mathbb{E}(XY)$ measures how strongly the two variables move together.

5.1 Autocorrelation function (ACF)

Autocorrelation function

If $\{X(t)\}$ is a random process, then its **autocorrelation function** (ACF) is defined by

$$R_{XX}(t_1, t_2) = \mathbb{E}[X(t_1) X(t_2)].$$

For a stationary random process $\{X(t)\}$, the ACF will be a function of the time difference $t_2 - t_1$ (or $t_1 - t_2$).

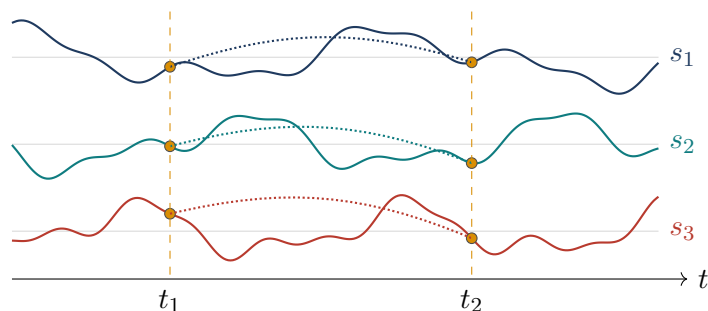


Figure 12: The ACF as an ensemble average of the products $X(t_1)X(t_2)$.

Multiply the pair of values on each sample function and **average over the ensemble**:

$$R_{XX}(t_1, t_2) \approx \frac{1}{N} \sum_{k=1}^N x(t_1, s_k) x(t_2, s_k).$$

5.2 Notation and the stationary case

The ACF is generally denoted by $R_{XX}(t_1, t_2)$ or $R_X(t_1, t_2)$ or $R(t_1, t_2)$, so that

$$R_X(t_1, t_2) = \mathbb{E}[X(t_1) X(t_2)].$$

For a stationary RP

$$R_X(t_1, t_2) = \phi(t_1 - t_2),$$

a **function of the time difference** $\tau = t_1 - t_2$ (or $t_2 - t_1$) only.

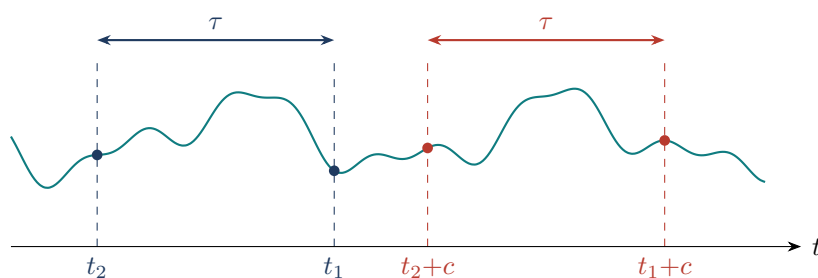


Figure 13: The same gap τ anywhere on the time axis gives the same value of R for a stationary random process.



Figure 14: Two typical autocorrelation functions: the peak occurs at $\tau = 0$, where $R(0) = \mathbb{E}[X^2(t)]$ is the mean power, and R is symmetric about $\tau = 0$.

5.3 Significance of the ACF

Note

The main significance of the ACF is that the **expected frequencies of the signal** in the next two time instants can be identified.

A slowly varying (low-frequency) signal stays correlated over a long time gap, so its ACF is broad; a rapidly varying (high-frequency) signal loses correlation quickly, so its ACF is narrow.

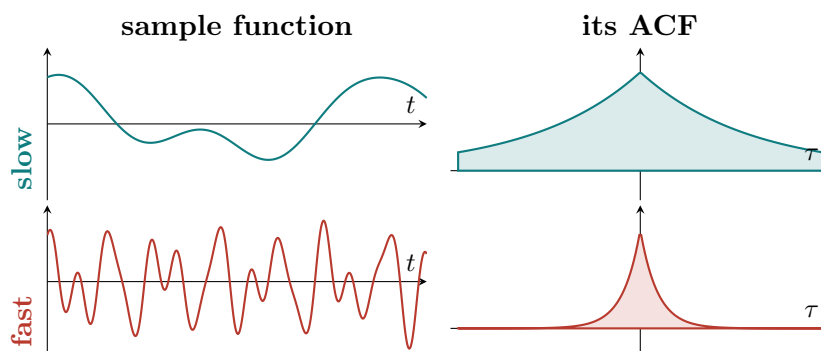


Figure 15: Slowly varying (low-frequency) signal $\Rightarrow$ broad ACF; rapidly varying (high-frequency) signal $\Rightarrow$ narrow ACF.

6 Wide-sense stationary (WSS) process

WSS process

A random process $\{X(t)\}$ is said to be a **wide-sense stationary** (WSS) process if

- (i) $\mathbb{E}[X(t)] = \text{a constant}$,
- (ii) the ACF $R_X(t_1, t_2)$ is a function of the time difference,
i.e. $R_X(t_1, t_2) = R_X(\tau)$ with $\tau = t_1 - t_2$.

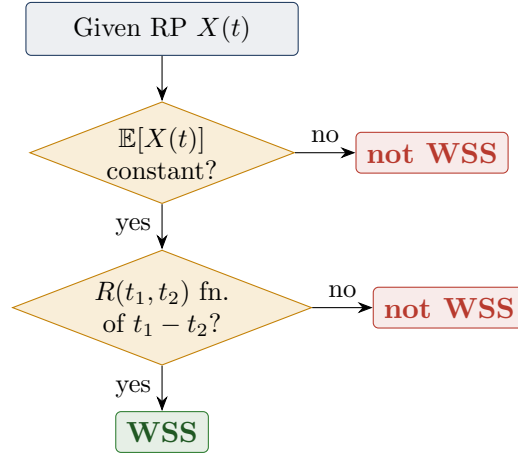


Figure 16: Testing a random process for wide-sense stationarity.

6.1 SSS and WSS side by side

SSS process

- $\mathbb{E}[X(t)] = \text{constant}$
- $\text{Var}[X(t)] = \text{constant}$

(formally: all joint distributions are shift-invariant)

WSS process

- $\mathbb{E}[X(t)] = \text{constant}$
- $R_X(t_1, t_2) = R_X(t_1 - t_2)$

(only first- and second-order moments)

For a WSS process the variance is automatically constant:

$$\text{Var}[X(t)] = \mathbb{E}[X^2(t)] - \mu^2 = R_X(0) - \mu^2.$$

All random processes

WSS

SSS

Figure 17: With finite second moments, every SSS process is also WSS; the converse need not hold.

6.2 Illustration: the random-phase cosine

Back to Example 2

Let $X(t) = A \cos(\omega t + \phi)$, where A and ω are constants and ϕ is uniformly distributed in $(0, 2\pi)$.

Mean.

$$\mathbb{E}[X(t)] = \frac{A}{2\pi} \int_0^{2\pi} \cos(\omega t + \phi) d\phi = 0 \quad (\text{a constant}).$$

Autocorrelation function.

$$\begin{aligned}
 R(t_1, t_2) &= A^2 \mathbb{E}[\cos(\omega t_1 + \phi) \cos(\omega t_2 + \phi)] \\
 &= \frac{A^2}{2} \mathbb{E}[\cos \omega(t_1 - t_2) + \cos(\omega(t_1 + t_2) + 2\phi)] = \frac{A^2}{2} \cos \omega(t_1 - t_2),
 \end{aligned}$$

which is a function of the time difference $\tau = t_1 - t_2$ alone. Hence $\{X(t)\}$ is a **WSS process**.

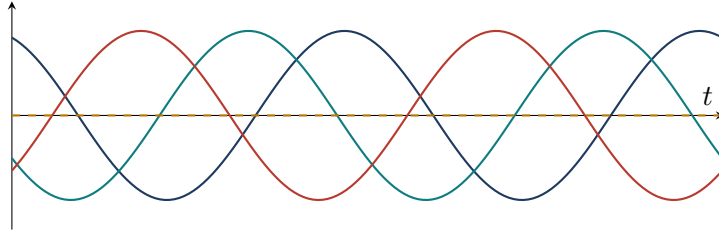


Figure 18: Sample functions of $X(t) = A \cos(\omega t + \phi)$ for three values of the random phase ϕ ; the ensemble mean (dashed) is zero at every t .

7 Summary

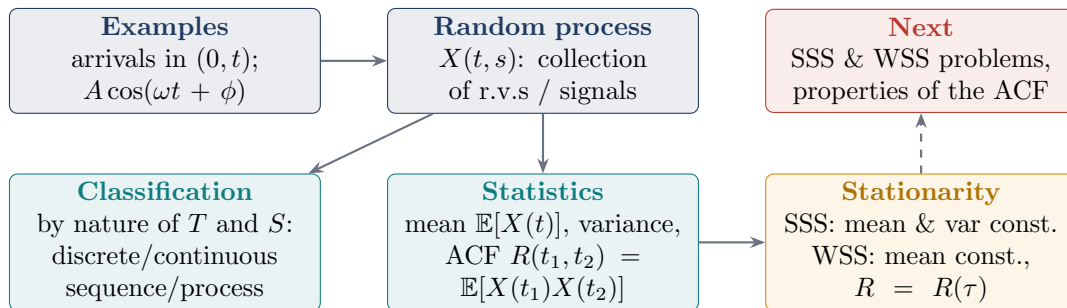


Figure 19: Concept map of Unit III – random process basics.

Quick revision

1. RP = collection of random variables; $X(t, s)$ involves both S and T .
2. T decides discrete / continuous; S decides sequence / process.
3. Stationary: statistical averages do not change with t .
4. SSS: mean and variance are constants.
5. ACF: $R_X(t_1, t_2) = \mathbb{E}[X(t_1)X(t_2)]$; for a stationary RP it depends only on $t_1 - t_2$.
6. WSS: mean is a constant *and* the ACF is a function of the time difference.